

**FH Aachen**

**Department of Electrical Engineering and Information Technology**

# **Exposé**

**Evaluating AI Agent Capabilities in SysML v2 Modeling  
via an MCP Bridge to Magic Systems of Systems Architect**

Nassim Awabdy

nassim.awabdy@alumni.fh-aachen.de

Supervisor: Prof. Dr. rer. nat. Sebastian Voss

Second examiner: M. Eng. Tim Höner

Master's Thesis – WS 2026/2027  
Aachen, September 2026

# 1 Exposé

Model-Based Systems Engineering (MBSE) manages the growing complexity of cyber-physical systems through a central, formalized system model [1]. SysML is the established language for such models. Its most recent revision, SysML v2, rebuilds the language on the formally grounded Kernel Modeling Language (KerML) [2], with a textual notation and a standardized API for tool access.

This thesis empirically evaluates the SysML v2 modeling capabilities of LLM-based agents operating interactively on a live model inside the industrial tool Magic Systems of Systems Architect (MSoSA). A purpose-built Model Context Protocol (MCP) bridge connects the agent to MSoSA's Java Open API (plugin API) and exposes the live model state as tools. Based on the failures observed with this bare bridge, it is then extended by a harness of knowledge base, guardrails and encoded best practices intended to improve adherence to modeling conventions. The open-source Airbus Apollo 11 SysML v2 model [3] serves as the system under test, with the GfSE model collection [4] as a smaller secondary set.

In a recent review of LLMs for system modeling [5], interactive, tool-using agents on live models do not appear as an evaluated subject. The closest work either evaluates file-based tooling only [6] or contributes an MCP framework without a capability evaluation [7], and no standard benchmark for fundamental modeling tasks exists [8, 9]. Closing this gap matters, as industry deploys agentic MBSE tooling faster than it is measured.

**RQ:** *Which SysML v2 modeling tasks can an LLM-based agent perform reliably in an industrial modeling environment (MSoSA) when connected through an MCP bridge, and where do its capabilities break down?*

The evaluation is organized as a benchmark of six use cases on Apollo 11: **Create**, **Query**, **Patch** and **Delete** as primitives on the live model, plus **Verification** and **Validation**, distinguished according to ISO/IEC/IEEE 15288 [10]. Each use case is instantiated in several difficulty tiers, from single operations to composed, realistic modeling flows such as fault correction, with faults injected into the model following Cibrián et al. [11] and Al-Shami et al. [12]. The evaluation proceeds in two phases. First, the agent is benchmarked through the bare MCP bridge to establish a baseline. The collected failure data then drives the design of the harness, and the benchmark is repeated to measure the improvement. Every task has a ground truth, and the resulting model state is scored by an evaluator outside the agent loop [13]. Tasks are repeated to account for LLM non-determinism [14]. Runs use Claude-family models. The outcome is a quantitative capability profile per use case and tier, before and after the harness, plus a qualitative failure taxonomy [15].

## Bibliography

- [1] INTERNATIONAL COUNCIL ON SYSTEMS ENGINEERING (INCOSE): Systems Engineering Vision 2020 / INCOSE. Version: September 2007. [https://sdincose.org/wp-content/uploads/2011/12/SEVision2020\\_20071003\\_v2\\_03.pdf](https://sdincose.org/wp-content/uploads/2011/12/SEVision2020_20071003_v2_03.pdf). 2007 (INCOSE-TP-2004-004-02, v2.03). – Technical Report. – INCOSE foresight report (workshop synthesis, not an empirical study) that coins the canonical MBSE definition. Anchors the thesis’s framing of MBSE and of the system model as the central artifact the agent operates on.
- [2] OBJECT MANAGEMENT GROUP: Kernel Modeling Language (KerML), Version 1.0 / Object Management Group. Version: March 2026. <https://www.omg.org/spec/KerML/1.0/PDF>. 2026 (formal/26-03-01). – Specification. – Normative OMG specification of the logic-grounded kernel SysML v2 extends, with four declarable tool conformance levels. Supplies the vocabulary for what an MCP bridge exposes – structural CRUD versus semantic conformance – and why v2 constraints are natively verifiable.
- [3] HELLE, Philipp ; SCHRAMM, Gerrit: Fly me to the Moon – Modeling Apollo 11 Using SysML v2. In: *Systems Engineering* (2026), June. <http://dx.doi.org/10.1002/sys.70074>. – DOI 10.1002/sys.70074. – [TOP 3] Airbus Central R&T’s five-layer CoSMA (Purpose/Operational/Functional/Logical/Technical) SysML v2 model of the Apollo 11 mission, released open-source ([github.com/airbus/apollo-11-sysml-v2](https://github.com/airbus/apollo-11-sysml-v2), MPL-2.0, commit 6e9c93f, accessed 20.09.2026) as a public MBSE benchmark. The instrument this thesis measures on: ~7.2 kLOC / 28 files / ~2 000 elements, deliberately incomplete “scaffold” for Create and fault correction tasks.
- [4] GESELLSCHAFT FÜR SYSTEMS ENGINEERING (GFSE): *SysML-v2-Models: An Open Collection of SysML v2 Models*. GitHub repository, <https://github.com/GfSE/SysML-v2-Models>, commit ebbb0c3, accessed 20.09.2026, 2025. – BSD-3. ~5.1 kLOC curated teaching/reference models (VehicleModel, DroneModelLogical, EveOnline Mining Frigate) explicitly meant to “help train LLMs”; ships `CI/parse.py` + a pilot-implementation jar as a validity oracle. Secondary set for generalisation: small self-contained models plus deliberate “bad examples”.
- [5] FRESEMANN, Carina ; ELLSEL, Claudius ; GRUNENWALD, Carl-Philipp ; STARK, Rainer: Using Large Language Models for System Modeling: Current State and Future Implications. In: *2025 IEEE International Symposium on Systems Engineering (ISSE)*, IEEE, October 2025, p. 1–8. – Structured review of 20 publications on LLMs for MBSE tasks (creation, analysis, modification, reformulation). Supplies the taxonomy behind this the-

sis's use-case catalogue, names MCP as the emerging context mechanism, and flags the missing evaluation criteria.

- [6] DEMPSEY, Colin: *Native AI Integration for Model-Based Systems Engineering: Three Layers that Make It Work*. SEI Blog, Carnegie Mellon University, <https://www.sei.cmu.edu/blog/native-ai-integration-for-model-based-systems-engineering-three-layers-that-make-it-work/>, September 2026. – [TOP 3] Controlled three-arm SysML v2 experiment (no tool / validation loop / plus MCP knowledge and skills): both tool arms reach zero syntax errors, but only full tooling lifts the pattern score (71.7 → 94.1). Direct template for this thesis's ablation arms; grey literature (practitioner blog), so its numbers are treated as unvalidated.
- [7] BAZZAL, Mahmoud ; SCHLEIFER, Simon ; LUNGU, Adriana ; KRUSE, Benjamin ; GOETZ, Stefan ; KOLDEWEY, Christian ; TISSEN, Denis ; WARTZACK, Sandro J. ; DUMITRESCU, Roman: Model Context Protocol for MBSE: A Unified Interoperable Agentic Framework for Systems Modelling. In: *Proceedings of the Design Society 6* (2026), July, p. 2721–2730. <http://dx.doi.org/10.1017/pds.2026.10630>. – DOI 10.1017/pds.2026.10630. – [TOP 3] MCP servers on the standard SysML v2 API with creation, modification and analysis tools, shown only through two qualitative case studies without metrics. Closest competing work: it delivers the artifact class, the thesis the missing capability evaluation inside MSoSA.
- [8] BOUAMRA, Yasmine ; YUN, Bruno ; POISSON, Alexandre ; ARMETTA, Frédéric: Sys-Temp: A Multi-Agent System for Template-Based Generation of SysML v2. In: *Advances in Practical Applications of Agents, Multi-Agent Systems, and Computational Social Science: The PAAMS Collection – 23rd International Conference, PAAMS 2025, Lille, France* Vol. 16031, Springer Nature Switzerland, June 2025 (Lecture Notes in Computer Science (LNAI)), p. 41–52. – Multi-agent template pipeline for NL-to-SysML v2 generation; scaffolding reaches zero syntax errors in 4/5 toy scenarios vs. 1/5 without, syntax only. The gap citation for this thesis's benchmark ("absence of standard benchmarks") and for SysML v2 corpus scarcity.
- [9] QUAST, Vincent ; JACOBS, Georg ; DEHN, Simon ; HÖPFNER, Gregor: Enabling Humans and AI Systems to Retrieve Information from System Architectures in Model-Based Systems Engineering. In: *Systems* 14 (2026), January, No. 1, p. 83. <http://dx.doi.org/10.3390/systems14010083>. – DOI 10.3390/systems14010083. – GraphRAG multi-agent QA over a Neo4j graph parsed from SysML v2 text, reaching 93% accuracy on a synthetic battery-EV model. Accuracy-metric template for Query and a second "no standardized benchmarks" gap citation; it queries a derived graph, not the live model.

- [10] ISO/IEC/IEEE: Systems and software engineering – System life cycle processes / International Organization for Standardization. Version: May 2023. <https://www.iso.org/standard/81702.html>. 2023 (ISO/IEC/IEEE 15288:2023). – International Standard. – Defines Verification (conformance to specified requirements) and Validation (fulfilment of stakeholder needs in the operational environment) as distinct life-cycle processes. The thesis adopts these definitions for its Verification and Validation use cases and audits prior work against them.
- [11] CIBRIÁN, Eduardo ; OLIVERT-ISERTE, Jose ; DÍEZ-FENOY, Carlos ; MENDIETA, Roy ; LLORENS, Juan ; ÁLVAREZ-RODRÍGUEZ, Jose M.: Ensuring Semantic Consistency in SysML v2 Models Through Metamodel-Driven Validation. In: *IEEE Access* 13 (2025), p. 121444–121457. <http://dx.doi.org/10.1109/ACCESS.2025.3587786>. – DOI 10.1109/ACCESS.2025.3587786. – Metamodel-derived rule checking (lexical, ANTLR syntax, semantic cross-references) detects 97 % of 428 injected errors across 25 OMG test models. Despite its title, ISO-verification rather than validation; precedent for Verification: an error taxonomy, a fault-injection method transferable to fault correction, and a detection-rate baseline.
- [12] AL-SHAMI, Haitham ; MALIK, Rohail ; ALA-LAURINHO, Riku ; VEPSÄLÄINEN, Jari ; VIITALA, Raine: Automated Semantic Fault Localization in SysML v2: A Human-in-the-Loop Framework Using Knowledge-Graph Augmented LLMs. In: *INCOSE International Symposium* 36 (2026), July, No. 1, p. 1274–1286. <http://dx.doi.org/10.1002/iis2.70189>. – DOI 10.1002/iis2.70189. – Fine-tuned small LLM plus a physical-compatibility KG repairs *semantic* faults that pass SysML v2 syntax checks, rising from <3 % to >91 % on 1 184 test samples. Methodological precedent for the fault injection and fault taxonomy of fault correction.
- [13] WANG, Zizhe: *Beyond Executable Models: The Pufibara Agent Harness and the Modelica Agent Workflow Benchmark for Physical System Modeling*. arXiv:2608.23653 [cs.SE], <https://arxiv.org/abs/2608.23653>, August 2026. – Agent harness plus a source-grounded 232-task Modelica benchmark (repair, generation, tuning) scored by an evaluator outside the agent loop. Supplies three adopted principles: external evaluator, tasks grounded in an existing model, and the repair/generation/tuning split.
- [14] PRADAS-GOMEZ, Alejandro ; BRAHMA, Arindam ; ISAKSSON, Ola: *DUCTILE: Agentic LLM Orchestration of Engineering Analysis in Product Development Practice*. arXiv:2603.10249, <https://arxiv.org/abs/2603.10249>, March 2026. – Splits adaptive orchestration by the agent from deterministic execution by verified tools, evaluated on an industrial aerospace task against expert acceptance criteria over 10 re-

peated runs. Template for the agent–tool division of labour and for repeated-run reliability reporting.

- [15] MOLINARI, Gioele ; FELTEN, Florian ; MASSOUDI, Soheyl ; FUGE, Mark: *EngiAI: Capability-Based Evaluation of Tool-Connected LLM Agents for Engineering Design*. arXiv:2605.19743 (extended version of an ASME IDETC-CIE 2026 paper), <https://arxiv.org/abs/2605.19743>, May 2026. – Capability-based evaluation scoring workflow execution, parameter selection, orchestration and code authoring separately from execution traces; backends reach 96–97 % on workflow tasks but only 20–53 % on tool-based decisions. Rationale for per-use-case scoring and template for the failure taxonomy.
- [16] BOELEN, Kathrin ; MAY, Matthias ; JACOBS, Georg ; MENNICKEN, Manuel ; MOERS, Frederik ; ZERWAS, Thilo ; HÖPFNER, Gregor: SysML v2 based modelling guidelines for mechanical system elements. In: *Forschung im Ingenieurwesen* 89 (2025), No. 1, p. 60. <http://dx.doi.org/10.1007/s10010-025-00827-w>. – DOI 10.1007/s10010-025-00827-w. – Encodes mechanical-domain modelling guidelines natively in SysML v2 via Metadata Definitions (moteego methodology, one worked example, no quantitative evaluation). Precedent for exposing guideline conformance as a tool-native check in Validation.
- [17] AHLBRECHT, Alexander ; ZAESKE, Wanja ; DURAK, Umut: MBSqlE: Enabling SQL as Powerful Query Language for SysML v2 in Aviation. In: *44th AIAA DATC/IEEE Digital Avionics Systems Conference (DASC 2025)*. Montreal, Canada : IEEE, September 2025. – The standard SysML v2 API offers no graph traversal (REST traversals >10 s); MBSqlE imports models into SQLite so recursive SQL traverses them in one query. Motivates a semantic/cache layer instead of a thin REST proxy for Query.
- [18] DEHART, John K.: Leveraging Large Language Models for Direct Interaction with SysML v2. In: *INCOSE International Symposium* 34 (2024), July, No. 1, p. 2168–2185. <http://dx.doi.org/10.1002/iis2.13262>. – DOI 10.1002/iis2.13262. – Founding paper of the LLM×SysML v2 topic: conceptual analysis and case studies, no controlled evaluation or ground truth. Provides the vision the thesis tests empirically and the “LLM edits text directly” pattern behind the no-tool arm.
- [19] DEHN, Simon ; SCHNÜRER, Simon ; JACOBS, Georg ; HÖPFNER, Gregor: Generating SysML v2 Models from Natural Language Requirements Using Large Language Models. In: *2025 IEEE International Symposium on Systems Engineering (ISSE)*, IEEE, October 2025, p. 1–7. – Four-component structured prompting for NL→SysML v2 generation, evaluated on an automotive use case with precision/recall/F1 and syntax checks

over three runs; few-shot examples prove essential. Primary reference for Create, but file-based without a modelling tool.

- [20] QUALIS, Richard A.: Mitigating Hallucinations in SysML v2 Generation Using LLMs and a Tri-Layered Knowledge Graph Reasoning Framework. In: *2025 ACM/IEEE 28th International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, IEEE, October 2025, p. 357–366. – Tri-layered knowledge graph grounding SysML v2 generation against hallucinations, evaluated on a satellite case study against curated ground truth: structurally valid, semantically weak. Motivates grounding the agent in the live model rather than a static KG (Create).
- [21] WU, Rui ; WEN, Guangwei ; ZHANG, XiaoNing ; LIANG, XiaoJun ; YANG, Chunhua ; GUI, Weihua: Knowledge-Augmented Conversational Agents for Dependency-Aware SysML v2 Model Generation and Refinement. In: *2025 IEEE 26th China Conference on System Simulation Technology and its Applications (CCSSTA)*, IEEE, July 2025, p. 581–586. – SysForge: four agents with dependency-aware KG retrieval and a generate–validate–refine loop, beating plain-LLM and semantic-RAG baselines on Pass@1 and BLEU. Closest prior work for fault correction, but file-based with its own validator instead of tool-native validation.
- [22] MOLNÁR, Vince ; GRAICS, Bence ; VÖRÖS, András ; TONETTA, Stefano ; CRISTOFOROTTI, Luca ; KIMBERLY, Greg ; DYER, Pamela ; GIAMMARCO, Kristin ; KOETHE, Manfred ; HESTER, John ; SMITH, Jamie ; GRIMM, Christoph: Towards the Formal Verification of SysML v2 Models. In: *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS Companion '24)*, 2024, p. 1086–1095. – OMG Formal Methods WG survey of five formal-methods integrations for SysML v2 (model checking, theorem proving, contracts), illustrated by small examples without benchmarks. State-of-the-art backdrop for positioning a prompt-driven agent's Verification capability, which has no formal backend.
- [23] RATZKE, Axel ; KOCH, Johannes ; GRIMM, Christoph: Modeling and Analysis of System Models with Constraints in SysMLv2. In: *2025 20th Annual System of Systems Engineering Conference (SoSE)*, 2025, p. 1–6. – Extends SysML v2 constraints with range semantics (`oneOf/anyOf/allOf`) compiled into a CSP, demonstrated on two small examples and limited to static structure. Mechanism reference for what an agent could delegate to a tool-native constraint solver in Verification.
- [24] LIPARULO, Leonardo ; PIERRI, Francesco: *Benchmarking AI Agents for Hardware Design Automation via MCP Tool Calling*. arXiv:2608.26199 [cs.AI], <https://arxiv.org/abs/2608.26199>, August 2026. – Benchmarks seven open-source models against a purpose-built MCP server for hardware design across single-operation,

chained, invalid and misspelled tasks under prompt and tool-description ablations. Closest methodological sibling: its task tiers and tool-description findings shape the bridge and benchmark design.

- [25] HASAN, Mohammed M. ; LI, Hao ; RAJBAHADUR, Gopi K. ; ADAMS, Bram ; HASSAN, Ahmed E.: *Model Context Protocol (MCP) Tool Descriptions Are Smelly! Towards Improving AI Agent Efficiency with Augmented MCP Tool Descriptions*. arXiv:2602.14878, <https://arxiv.org/abs/2602.14878>, February 2026. – Study of 856 MCP tools: 97.1 % of descriptions carry a “smell”, and augmenting them raises success by 5.85 pp but adds 67 % more steps. Supplies the rubric for the bridge’s tool descriptions and a confound to hold constant across arms.
- [26] SHI, Jiasheng ; ZHANG, Tianhan: *What Do CAE Simulation Agents Really Need Beyond a Generic Harness?* arXiv:2609.03718 [cs.CE], <https://arxiv.org/abs/2609.03718>, September 2026. – With information access and repair budget held fixed, a generic single-agent harness beats specialised multi-agent CAE systems (96.4 % vs. 88.2 %); execution-feedback repair and domain tutorials drive the gains. Counter-evidence that the thin-vs-full-harness ablation must engage.
- [27] SHEFA, Jannatul ; SALADO, Alejandro ; WACH, Paul ; TOPCU, Taylan G.: *Two Truths and A Lie? Benchmarking Off-the-Shelf LLMs for Requirements Quality Assessment: Performance, False Alarms, and Misses*. arXiv:2609.03230 [cs.SE], <https://arxiv.org/abs/2609.03230>, September 2026. – Ten LLMs, 100 runs, expert ground truth on INCOSE criteria: the best model finds a median of 47 % of issues and misses judgement-heavy ones. Justifies expert acceptance criteria with a human second rater instead of LLM-as-judge, and repeated runs.